

Geometric Look-Angle Shaping Strategy for Enclosed Inspection

Amit Shivam¹, Manuel C.R.M. Fernandes¹, Sérgio Vinha¹, and Fernando A.C.C. Fontes¹

Abstract—This paper introduces inspection through GLASS, a Geometric Look-Angle Shaping Strategy for enclosed regions using unmanned aerial vehicles. In doing so, the vehicle’s guidance command is constructed through a bounded, look-angle shaping function relative to a desired standoff path. By embedding a smooth, hyperbolic-tangent-type shaping function within a polar geometric framework, GLASS ensures global existence of the guidance dynamics while avoiding the far-field limitations inherent to conventional formulations. Lyapunov stability analysis establishes asymptotic convergence to a prescribed inspection standoff under explicit curvature feasibility conditions, along with analytical settling-time characteristics. The proposed strategy incorporates maximum turn-rate constraints without inducing singularities throughout the workspace. High-fidelity six-degree-of-freedom quadrotor simulations demonstrate the effectiveness of GLASS in representative enclosed inspection scenarios, highlighting a practically viable guidance framework for autonomous enclosed inspection missions.

I. INTRODUCTION

Unmanned aerial vehicles (UAVs) are increasingly employed in persistent surveillance, reconnaissance, and cooperative monitoring missions, where a UAV is often required to regulate its motion relative to a target to maintain a prescribed radial separation, commonly referred to as the stand-off distance. Such missions require guidance strategies that can precisely control the relative geometry between the UAV and the target, particularly under the kinematic and curvature constraints inherent to fixed-wing platforms. Moreover, real-world mission requirements impose time-critical performance objectives, necessitating guidance strategies that are computationally efficient, structurally simple, and capable of providing deterministic convergence guarantees.

Existing standoff guidance methods can broadly be classified into virtual-target-based approaches [1], [2], relative side-bearing pursuit methods [3], bifurcation-based strategies [4]–[7], and vector-field guidance approaches [8]–[15]. Virtual-target and pursuit-based methods [1]–[3] offer structurally simple guidance laws, but their transient performance may deteriorate under large initial separation and often induce high turn-rate demand during far-field capture.

Bifurcation-based formulations [4]–[7] reshape the engagement dynamics through tunable bifurcation parameters to realize desired motion primitives and inspection behaviors. However, their convergence characteristics remain implicitly embedded within the design parameters, limiting explicit performance characterization.

Vector-field methods provide position-based heading guidance commands that eventually drive the UAV towards the desired path. Classical Lyapunov vector fields [8] combine circulation and contraction components to achieve asymptotic convergence, while subsequent works [9]–[11] improved transient behaviour and curvature characteristics through modified vector-field guidance methods. More recently, arcsine-based shaping approaches [12]–[15] constructed globally defined heading guidance commands for circular, elliptic, and Lamé-curve-based path following. Although these methods provide analytical stability guarantees, their convergence behavior remains strongly coupled to the chosen shaping function, leading to implicit trade-offs between radial convergence speed, curvature demand, and transient turn-rate peaks. Jha *et al.* [16] introduced an arcsine-based look-angle shaping framework with closed-form settling-time characterization and explicit gain bounds. Despite providing deterministic convergence guarantees, the feasible gain becomes increasingly restricted for large initial separation to preserve real-valued look-angle dynamics, thereby limiting achievable far-field convergence rates under turn-rate constraints.

Motivated by these limitations, this work develops a geometry-driven look-angle shaping framework in which the approach geometry itself acts as the primary design variable. Instead of enforcing faster convergence through gain amplification, the proposed method prescribes a bounded nonlinear evolution of the look angle that intrinsically regulates radial contraction and curvature demand. Constructed directly in polar coordinates with explicit turn-rate feasibility embedded into the design, the resulting closed-loop dynamics admit analytically tractable settling-time characterization under radial-dominant conditions. Consequently, the proposed framework enables rapid far-field capture while preserving curvature feasibility, thereby enlarging the admissible operational envelope for practical standoff tracking missions.

The remainder of the paper is organized as follows. Section II presents the problem formulation. Section III develops the proposed geometry-driven look-angle shaping guidance law. Simulation studies and comparative performance evaluation are presented in Section IV. Finally, concluding remarks and future research directions are discussed in Section V.

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II. PROBLEM FORMULATION

Consider a problem scenario as shown in Fig. 1, where a UAV is required to inspect a planar enclosed region from a standoff distance r_d . The centre of the region is assumed to be located at the origin of an inertial frame. For a planar

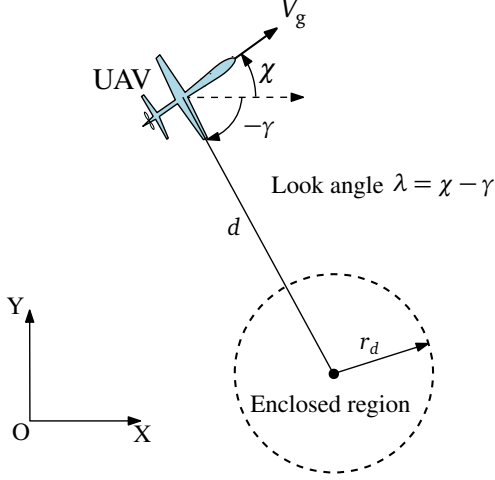


Fig. 1: Problem scenario

inspection geometry, d and γ denote the range and the line-of-sight angle (LOS) on the plane of the UAV's trajectory relative to the origin. The look angle λ is the difference between the instantaneous course and the line-of-sight angle, which is expressed as

$$\lambda = \chi - \gamma. \quad (1)$$

The relative engagement kinematics of the UAV in polar coordinates are described by

$$\begin{aligned} \dot{d} &= V_g \cos \lambda \\ \dot{\gamma} &= \frac{V_g}{d} \sin \lambda, \end{aligned} \quad (2)$$

where V_g is the vehicle ground speed. Let the desired standoff path be specified in polar form as a smooth closed curve

$$d_{\text{des}} = r(\gamma), \quad (3)$$

where $r(\gamma)$ corresponds to a circle, ellipse, or Lamé (superellipse) curve. The objective of this work is to design a guidance law for χ such that the standoff error reduces to zero and satisfies

- 1) The settling time to a prescribed tube standoff error $e = d - r(\gamma)$ such that $|e| \leq \varepsilon$ is analytically tractable.
- 2) The commanded course rate satisfies the actuator constraint

$$|\dot{\chi}| \leq \omega_{\max}, \quad (4)$$

where $\omega_{\max}$ is the maximum achievable yaw (or course) rate of the UAV.

III. PROPOSED LOOK-ANGLE BASED GUIDANCE LAW

This section discusses the geometry-driven proposed guidance law while emphasising the limitations of arcsine-based guidance law.

A. Far-Field Limitation of Arcsine Shaping Function

Look-angle shaping guidance laws used in [16] impose an intrinsic feasibility constraint on the shaping gain. For a representative arcsine construction of the form

$$\lambda_a(e) = \frac{\pi}{2} + \sin^{-1}(k_D e), \quad |k_D e(t)| < 1 \quad (5)$$

A necessary feasibility condition at $t = 0$ is $|k_D e(t)| < 1$, i.e., $k_D < \frac{1}{|e_0|}$. For far-field cases this forces k_D to be small, and therefore limits the achievable initial radial contraction.

B. Proposed GLASS Shaping Law

To overcome the above limitation, we propose a bounded hyperbolic-tangent shaping guidance as

$$\sigma(e) = -\tanh(k_G e), \quad k_G > 0, \quad (6)$$

embedded implicitly within the look-angle equation. Since $|\tanh(k_G e)| < 1$ for all $e \in \mathbb{R}$, the shaping function is globally defined and smooth for any combination of k_G and e , with uniformly bounded slope

$$\left| \frac{d\sigma}{de} \right| = k_G \operatorname{sech}^2(k_G e) \leq k_G. \quad (7)$$

The highlight of this formulation is that the hyperbolic-tangent shaping guidance does not suffer from the limitation imposed by the initial conditions. Using (3), define the geometry coupling

$$a(\gamma, d) = \frac{r'(\gamma)}{d}. \quad (8)$$

The commanded look angle λ is selected to satisfy the implicit GLASS constraint

$$\cos \lambda - a(\gamma, d) \sin \lambda = -\sigma(e). \quad (9)$$

C. Geometric Feasibility and Regularity

Theorem 1 (Global feasibility): Define

$$\alpha := \tan^{-1}(a), \quad (10)$$

$$\gamma_\lambda(e) := \cos^{-1}\left(-\frac{\sigma(e)}{\sqrt{1+a^2}}\right) \in [0, \pi]. \quad (11)$$

For any $e \in \mathbb{R}$, the algebraic constraint (9) admits a real solution $\lambda(e, \gamma, d)$ and the two solution branches

$$\lambda_d(e) = -\phi + s \gamma_\lambda(e), \quad s \in \{+1, -1\}. \quad (12)$$

Then the following hold:

- 1) **Existence for all** $e \in \mathbb{R}$. The quantity $\gamma_\lambda(e)$ is well-defined for all $e \in \mathbb{R}$, i.e.,

$$-\frac{\sigma(e)}{\sqrt{1+a^2}} \in (-1, 1) \quad \forall e \in \mathbb{R}, \quad (13)$$

and hence $\lambda_d(e)$ exists globally without any initial-condition-dependent feasibility constraint.

- 2) **Boundedness and smoothness.** Each branch $\lambda_d(e)$ is continuous for all $e \in \mathbb{R}$ and is continuously differentiable with bounded slope. In particular, for all $e \in \mathbb{R}$,

$$\left| \frac{d\lambda_d}{de} \right| = \left| \frac{d\gamma_\lambda}{de} \right| \leq \frac{k_G}{\sqrt{1+a^2}}. \quad (14)$$

Moreover, each solution branch is continuous and continuously differentiable in e , with globally bounded slope.

Proof: Using (9) and (10), the constraint becomes

$$\cos(\lambda + \alpha) = \frac{-\sigma(e)}{\sqrt{1+a^2}}. \quad (15)$$

Because $|\sigma(e)| < 1$ and $\sqrt{1+a^2} \geq 1$, the argument of $\cos^{-1}(\cdot)$ always lies strictly in $(-1, 1)$. Hence, a real solution exists for all $e \in \mathbb{R}$, independent of initial conditions. Differentiating (11) gives

$$\frac{d\gamma_\lambda}{de} = -\frac{1}{\sqrt{1 - (\sigma(e)/\sqrt{1+a^2})^2}} \cdot \frac{1}{\sqrt{1+a^2}} \cdot \frac{d\sigma}{de}. \quad (16)$$

Because $|\sigma(e)| < 1$ and $\sqrt{1+a^2} \geq 1$, we have

$$\begin{aligned} 0 &< 1 - \left(\frac{\sigma(e)}{\sqrt{1+a^2}} \right)^2 \leq 1 \\ \Rightarrow \frac{1}{\sqrt{1 - (\sigma(e)/\sqrt{1+a^2})^2}} &\leq \frac{1}{\sqrt{1-0}} = 1, \end{aligned} \quad (17)$$

and therefore, combining with (7) in (16) yields the global bound

$$\left| \frac{d\gamma_\lambda}{de} \right| \leq \frac{1}{\sqrt{1+a^2}} \left| \frac{d\sigma}{de} \right| \leq \frac{k}{\sqrt{1+a^2}}. \quad (18)$$

Since $\lambda_d(e) = -\phi + s\gamma_\lambda(e)$, the same bound holds for $d\lambda_d/de$, proving (14). Since $\sigma(e)$ is smooth with bounded derivative, and the denominator in the implicit derivative expression never vanishes, the slope $d\lambda/de$ remains globally bounded. Therefore, no arcsine-type singularity or initial-condition-dependent feasibility constraint arises. ■

D. Closed-Loop Stability on General Standoff Curves

Define the scalar standoff error

$$e(t) = d(t) - r(\gamma(t)). \quad (19)$$

Differentiating and substituting the polar engagement kinematics yields

$$\dot{e} = V_g (\cos \lambda - a(\gamma, d) \sin \lambda). \quad (20)$$

Imposing the GLASS constraint (9) gives the reduced scalar dynamics

$$\dot{e} = -V_g \tanh(k_G e). \quad (21)$$

In the ideal guidance setting, we assume the commanded look angle λ satisfies the algebraic GLASS constraint at all times, yielding the reduced scalar error dynamics below.

Theorem 2 (Global asymptotic convergence): Consider the scalar dynamics (21), then

- 1) Solutions exist uniquely for all $t \geq 0$.
- 2) The equilibrium $e = 0$ is globally asymptotically stable.
- 3) The error evolves monotonically toward zero without overshoot.

Proof: The right-hand side is globally Lipschitz because

$$\left| \frac{d}{de} (-V_g \tanh(k_G e)) \right| = V_g k_G \operatorname{sech}^2(k_G e) \leq V_g k_G. \quad (22)$$

Hence, solutions exist uniquely for all time. Consider the Lyapunov function

$$V(e) = \int_0^e \tanh(k_G s) ds = \frac{1}{k_G} \ln(\cosh(k_G e)). \quad (23)$$

Since $\cosh(\cdot) \geq 1$, we have positive definite $V(e) \geq 0$ with equality only at $e = 0$. Along trajectories,

$$\dot{V} = \tanh(k_G e) \dot{e} = -V_g \tanh^2(k_G e) \leq 0, \quad (24)$$

and $\dot{V} = 0$ only when $e = 0$. Thus, $e = 0$ is globally asymptotically stable. Furthermore, if $e > 0$ then $\dot{e} < 0$, and if $e < 0$ then $\dot{e} > 0$. Hence the solution cannot change sign and converges monotonically to zero. ■

E. Analytical Tube-Entry Time

Theorem 3: Under the hypotheses of Theorem 2, let $e(t)$ evolve according to (21) with initial error $e(0) = e_0$. For a prescribed tube size $\varepsilon > 0$, define the tube-entry time

$$T_\varepsilon \triangleq \inf\{t \geq 0 : |e(t)| \leq \varepsilon\}. \quad (25)$$

If $|e_0| > \varepsilon$, then the tube-entry time is given in closed form by

$$T_\varepsilon = \frac{1}{k_G V_g} \ln \left(\frac{\sinh(k_G |e_0|)}{\sinh(k_G \varepsilon)} \right). \quad (26)$$

If $|e_0| \leq \varepsilon$, then $T_\varepsilon = 0$.

Proof: From (21), the dynamics are odd and sign-preserving; thus it suffices to consider $e_0 > 0$ (the case $e_0 < 0$ follows by symmetry and yields the same expression in terms of $|e_0|$). For $e > 0$, (21) implies $\dot{e} < 0$, so $e(t)$ decreases monotonically until it reaches ε . Separation of variables for $e(t) \in [\varepsilon, e_0]$ yields

$$\frac{de}{\tanh(k_G e)} = -V_g dt. \quad (27)$$

Integrating from $t = 0$ to $t = T_\varepsilon$ and from $e(0) = e_0$ to $e(T_\varepsilon) = \varepsilon$ gives

$$\int_{e_0}^{\varepsilon} \frac{de}{\tanh(k_G e)} = -V_g \int_0^{T_\varepsilon} dt = -V_g T_\varepsilon. \quad (28)$$

Using the identity (valid for $e > 0$)

$$\int \frac{de}{\tanh(k_G e)} = \frac{1}{k_G} \ln(\sinh(k_G e)) + C,$$

we obtain

$$\begin{aligned} \int_{e_0}^{\varepsilon} \frac{de}{\tanh(k_G e)} &= \frac{1}{k_G} [\ln \sinh(k_G \varepsilon) - \ln \sinh(k_G e_0)] \\ &= -\frac{1}{k_G} \ln \left(\frac{\sinh(k_G e_0)}{\sinh(k_G \varepsilon)} \right). \end{aligned} \quad (29)$$

Substituting into (28) yields

$$-V_g T_\varepsilon = -\frac{1}{k_G} \ln \left(\frac{\sinh(k_G e_0)}{\sinh(k_G \varepsilon)} \right),$$

and hence

$$T_\varepsilon = \frac{1}{k_G V_g} \ln \left(\frac{\sinh(k_G e_0)}{\sinh(k_G \varepsilon)} \right).$$

Replacing e_0 by $|e_0|$ to cover both signs gives (26). If $|e_0| \leq \epsilon$, then the initial condition is already inside the tube, so $T_\epsilon = 0$ by definition (25). ■

IV. SIMULATION RESULTS

This section presents numerical simulation studies to evaluate the performance of the inspection strategy using the proposed GLASS-based guidance scheme. We first consider ideal guidance dynamics to examine the convergence properties of the geometric look-angle shaping mechanism. The analysis is then extended to a first-order course-control while satisfying turn-rate constraints. Finally, comparative simulations are conducted against representative guidance laws from the existing literature to highlight the advantages of the proposed approach. Unless stated otherwise, $V_g = 20$ m/s, a desired circular orbit with centre at the origin and radius $r_d = 200$ m, standoff settling tube $\epsilon = 0.05$ is considered. The saturation operator $\text{sat}(\cdot)$ is defined as $\text{sat}(x) = \max(\min(x, x_{\max}), -x_{\max})$, where $x_{\max}$ denotes the prescribed saturation bound. Further, the angular wrapping operator $\text{wrap}(\cdot)$ maps look angle to the interval $[-\pi, \pi)$ to avoid discontinuities due to angle periodicity.

A. Ideal dynamics

In the first case, numerical simulations are performed under ideal guidance dynamics, with $\chi = \chi_d$. Table I lists the initial conditions used for simulation for two cases: initial position outside and inside the desired orbit.

TABLE I: Initial conditions for ideal dynamics

Parameter	Symbol	Value	
		Outside	Inside
Initial position	(x_0, y_0)	(450, -250) m	(45, -25) m
Initial range	d_0	514.78 m	51.48 m
Initial standoff error	e_0	314.78 m	-148.52 m
Initial LOS angle	γ_0	-29.05°	-29.05°

Fig. 2 illustrates the simulation results for outside initial positions. Fig. 2a plots all UAV trajectories that converge smoothly toward the desired circular orbit. The corresponding standoff error profiles reducing to negligibly low value are shown in Fig. 2b. As indicated in Fig. 2b, increasing k_G values makes the radial error reaching to the prescribed ϵ tube, with larger shaping gains yielding shorter tube-entry times. Fig. 2c depicts the turn-rate profiles associated with the convergence maneuver. Larger k_G generate higher turn-rate peaks, resulting in large maximum curvature, as evident in Fig. 2d. Similarly, Fig. 3 plots numerical results for inside initial positions. UAV trajectories are plotted in Fig. 3a with the corresponding standoff error profiles reducing to zero is shown in Fig. 3b. The turn-rate profiles are shown in Fig. 3c, where increasing k_G results in turn rate peaking behaviour. Notably, the curvature–standoff error relationship in Fig. 3d remains smooth and bounded across all cases. For both outside and inside initial conditions, the numerically obtained settling times for the ideal guidance dynamics are listed in Table II.

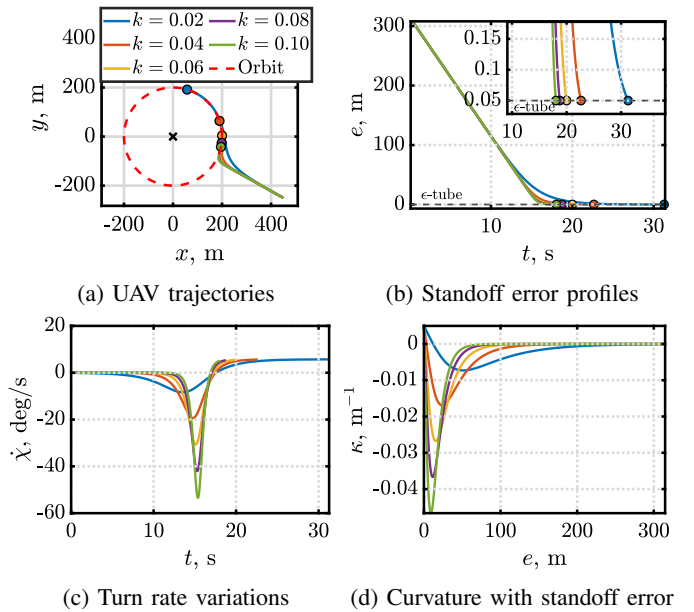


Fig. 2: Results for ideal dynamics for outside initial position

TABLE II: Closed-form analytic settling time T_s for ideal-manifold dynamics under inside and outside initial conditions for different gain values k_G .

Standoff	$k_G = 0.02$	$k_G = 0.04$	$k_G = 0.06$	$k_G = 0.08$	$k_G = 0.10$
Inside ($e_0 < 0$)	22.956	14.328	11.689	10.444	9.729
Outside ($e_0 > 0$)	31.276	22.641	20.002	18.757	18.042

B. Simulation with nonideal initial heading

In the second case, the simulation is carried out with a first-order course-angle dynamics governed by

$$\dot{\chi} = k_\chi(\chi_d - \chi) \quad (30)$$

where k_χ denotes the controller gain. The design and control parameters used in the simulation are $(k_G, k_\chi, \dot{\chi}_{\max}) = (0.02, 50, 0.5 \text{ rad/s})$, respectively. Fig. 4 presents the results for outside initial positions with both ideal and non-ideal initial heading offsets. Fig. 4a plots trajectories converging to the desired orbit despite significant heading mismatches (e.g., $-60^\circ, +90^\circ$) demonstrating the robustness of the GLASS guidance law to initial orientation errors. The corresponding error profiles are plotted in Fig. 4b and exhibit monotonic convergence to the ϵ -tube, with only a slight delay relative to the ideal-heading case due to finite course-loop bandwidth. The turn rate profiles are shown in Fig. 4c with the corresponding curvature–standoff error relationship plotted in Fig. 4d. Fig. 5 shows the corresponding results for inside initial positions. Even with non-ideal initial heading offsets, all trajectories smoothly converge to the desired orbit, as seen in Fig. 5a. The standoff error profiles in Fig. 5b remain monotonically decaying to zero and enter the prescribed ϵ -tube, with minor settling-time differences relative to the ideal-heading case. The corresponding turn rate profile and

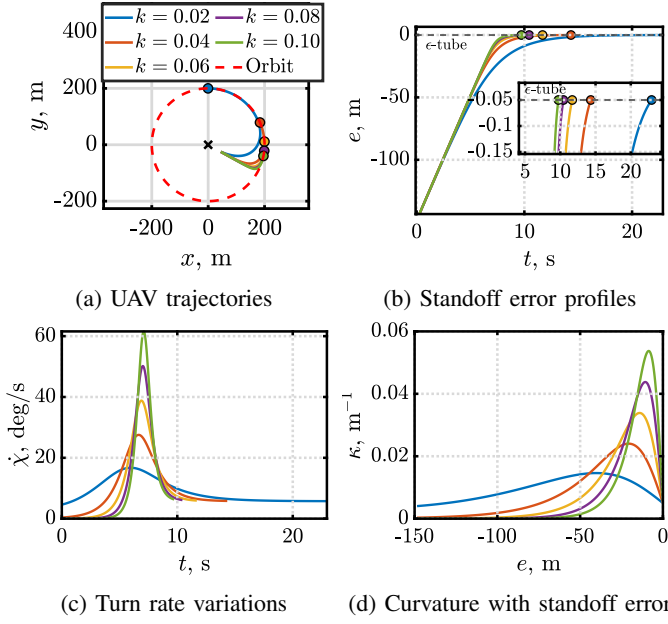


Fig. 3: Results for ideal dynamics for inside initial position

curvature variations with standoff errors are depicted in Fig. 5c and 5d. These results confirm that the geometric look-angle shaping mechanism maintains stability and bounded curvature demands even when both heading mismatch and finite course dynamics are present.

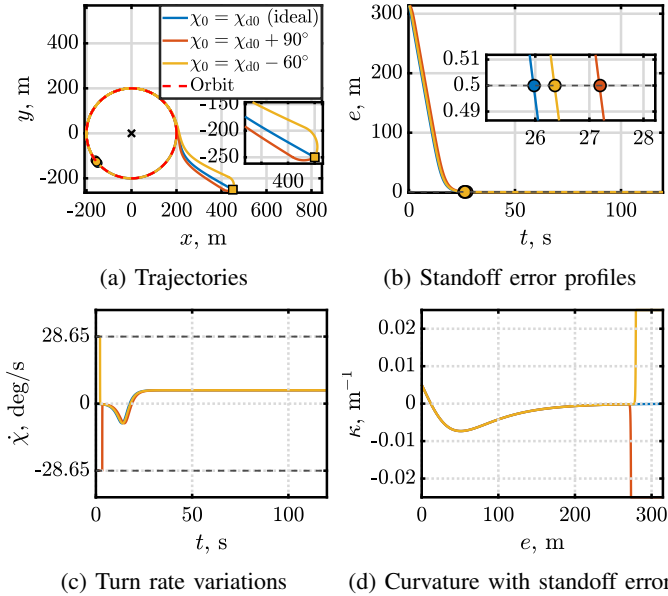


Fig. 4: Results for outside initial UAV position

Table III summarizes the initial conditions and corresponding analytical and simulated settling times for both outside and inside initial positions under varying heading mismatches. For each case, the analytical settling time T_{ana} is computed using the reduced scalar error dynamics (21) whereas the simulated time T_{sim} is obtained from full non-

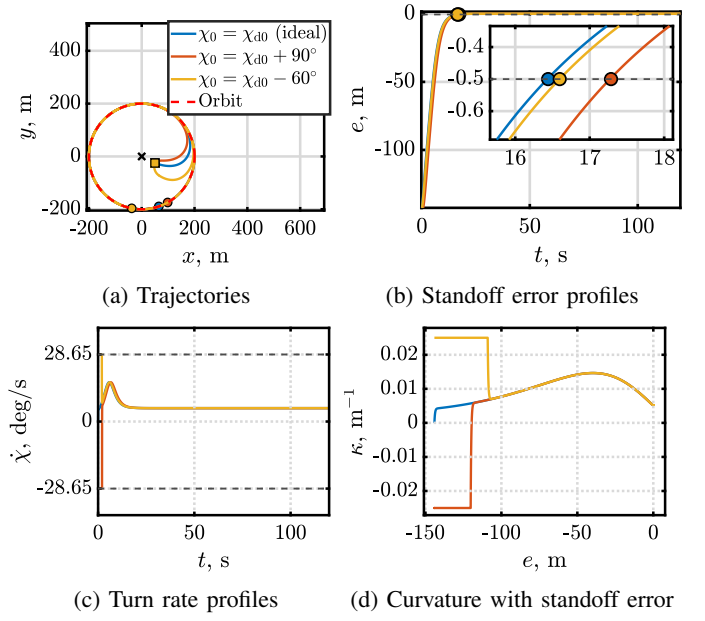


Fig. 5: Results for inside initial UAV position

linear simulations including heading dynamics and actuator constraints. For both outside and inside initial conditions, the analytical settling time remains constant for a given initial radial error, as expected from the closed-form expression which depends solely on (e_0, k_G, V_g) and is independent of initial heading mismatch. In contrast, the simulated settling time varies with the initial heading angle. This discrepancy arises because the analytical model assumes ideal heading tracking and immediate satisfaction of the algebraic GLASS constraint, whereas the simulation includes transient heading alignment dynamics governed by finite turn-rate limits. In the outside cases, larger heading mismatches introduce an additional alignment phase before the radial-dominant dynamics become effective, resulting in simulated settling times slightly exceeding the analytical prediction. The deviation remains modest, confirming that the reduced scalar model captures the dominant convergence behavior. Similarly, in the inside cases, the differences between analytical and simulated times are small, indicating that the GLASS guidance rapidly enforces the desired radial contraction once the heading error is corrected. Overall, the closeness between analytical and simulated settling times validates the theoretical reduction to scalar error dynamics, while the observed differences are primarily attributable to non-ideal heading transients and actuator rate limitations not accounted for in the analytical derivation.

C. Comparative study

To ensure a fair comparison between the proposed GLASS strategy and the arcsine-based look-angle guidance [16], the initial commanded heading angle is matched for both methods while satisfying the same maximum turn rate limits of 28-1 deg/s. Since the arcsine formulation directly prescribes

TABLE III: Initial conditions and settling times for heading mismatch

Case	(x_0, y_0) , m	χ_0 , deg.	λ_0 , deg.	T_{ana} , s	T_{sim} , s
Outside	(450, -250)	150.73	179.79	25.519	25.978
	(450, -250)	-119.27	-90.21	25.519	26.363
	(450, -250)	90.73	119.79	25.519	27.199
Inside	(50, -25)	-20.15	6.41	16.977	16.442
	(50, -25)	69.85	96.41	16.977	17.287
	(50, -25)	-80.15	-53.59	16.977	16.598

the look angle as

$$\sigma_D = \psi - \gamma = \frac{\pi}{2} + \sin^{-1}(k_D e_D),$$

where e_D is the normalized range error defined in Eq. (6) of [16], equality of the initial heading reduces to enforcing $\sigma_D(0) = \sigma_G(0)$, where $\sigma_G(0)$ is the LOS-measured look angle generated by the proposed method. For the circular standoff case, the GLASS constraint yields

$$\cos \sigma_G = -\tanh(k_G(d_0 - r_d)),$$

which uniquely determines the initial look angle for a selected orbit direction. Equating the two look angles gives the explicit gain mapping

$$k_D = \frac{\sin(\sigma_G(0) - \frac{\pi}{2})}{e_{0,D}},$$

where $e_{0,D} = -(1 - r_d/d_0)$ for $d_0 > r_d$. For the representative case $r_d = 200$ m and $d_0 = 650$ m (so that $d_0 - r_d = 450$ m) with $k_G = 0.007$, the GLASS shaping term evaluates to $\tanh(3.15) \approx 0.9963$, yielding $e_{0,D} = -9/13 \approx -0.6923$ and the matched arcsine gain $k_D \approx -1.439$. This guarantees identical initial look angles and hence identical initial headings for both guidance laws, ensuring that any subsequent differences in convergence rate, curvature demand, or saturation behavior arise solely from the intrinsic structure of the guidance design, as illustrated by the simulation results presented below. Figure 6 presents comparative simulation results for an outside initial UAV position. As shown in Fig. 6a, both guidance laws eventually converge to the desired standoff circle. However, the transient behavior differs markedly. This behavior is consistent with the far-field analysis, wherein the proposed hyperbolic-tangent shaping preserves a significant radial component even for large initial errors. Figure 6b compares the evolution of standoff error profiles. The GLASS strategy exhibits faster radial-error decay, reaching the ϵ -tube significantly earlier than the arcsine-based method. The turn-rate profiles in Fig. 6c demonstrate that the improved convergence speed of GLASS does not come at the cost of excessive control effort. Figure 6d further illustrates the curvature characteristics as a function of the standoff error.

D. 6DOF simulation results

We validate the GLASS guidance scheme-to-6DOF embedding on the dynamic quadrotor model [17] to assess the practicality of the proposed method in realistic applications.

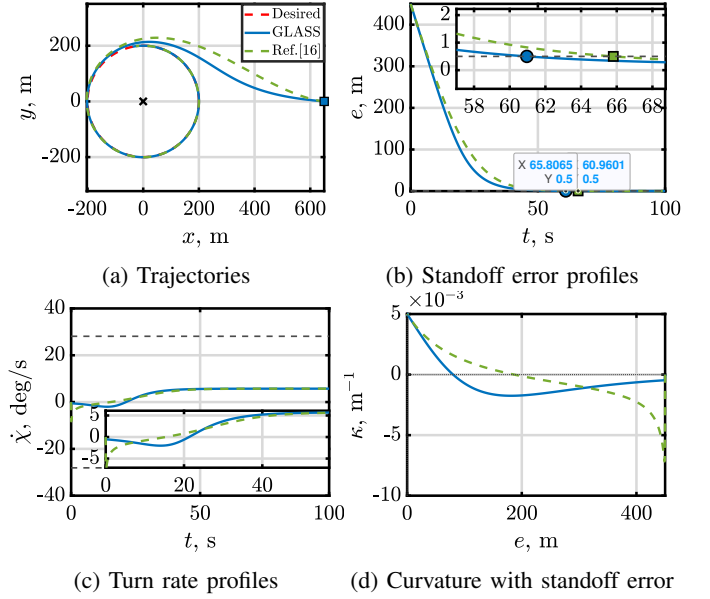


Fig. 6: Comparative results for outside initial UAV position

A quadrotor model cannot instantaneously realize $\chi = \chi_d$. We therefore introduce a first-order course autopilot with rate saturation:

$$\dot{\chi} = \text{sat}(\dot{\chi}_f(\chi_f - \chi), \omega_{\max}), \quad (31)$$

where χ_f is a filtered version of χ_d to mitigate oscillations induced by aggressive shaping gains:

$$\dot{\chi}_f = \frac{1}{\tau_{\chi_d}} \text{wrap}(\chi_d - \chi_f). \quad (32)$$

1) *Course-to-acceleration mapping*: Unlike fixed-wing kinematics, the OS4 quadrotor generates planar accelerations through thrust tilting. We convert the (filtered) course state χ into a horizontal velocity reference:

$$\hat{t} = \begin{bmatrix} \cos \chi \\ \sin \chi \end{bmatrix}, \quad \hat{r} = \begin{bmatrix} \cos \gamma \\ \sin \gamma \end{bmatrix}, \quad v_{\text{ref}} = V_{\text{ref}} \hat{t} - k_{\text{rad}} e \hat{r}. \quad (33)$$

The second term provides explicit radial correction and compensates for inner-loop lag and saturation, thereby reducing steady bias in the achieved orbit radius. A proportional velocity controller generates the commanded horizontal acceleration

$$a_{xy}^{\text{cmd}} = k_v (v_{\text{ref}} - v_{xy}), \quad \|a_{xy}^{\text{cmd}}\| \leq a_{\max}. \quad (34)$$

2) *Acceleration-to-attitude mapping and yaw synchronization*: Using the standard small-to-moderate tilt approximation, the commanded inertial accelerations (a_x, a_y) are mapped into desired pitch θ_d and roll ϕ_d commands using the instantaneous yaw ψ

$$\theta_d = \frac{a_x \cos \psi + a_y \sin \psi}{g}, \quad \phi_d = \frac{a_x \sin \psi - a_y \cos \psi}{g}, \quad (35)$$

with saturations $|\phi_d| \leq \phi_{\max}$ and $|\theta_d| \leq \theta_{\max}$. The yaw command is chosen consistent with the course channel,

$$\psi_d = \chi, \quad (36)$$

ensuring that the acceleration-to-attitude mapping remains coherent. Ref. [17] can be referred for more details. The simulation parameters are considered as $m = 0.24$ kg, $(I_x, I_y, I_z) = (2.3, 2.3, 4.0) \times 10^{-3}$ kg·m², $l = 0.20$ m, rotor limits $\Omega_i \in [0, 600]$ rad/s, and actuator mapping coefficients $(b, d, J_r) = (3.0 \times 10^{-6}, 1.0 \times 10^{-7}, 2.0 \times 10^{-5})$. The desired standoff radius is $r_d = 20$ m around $(x_c, y_c) = (0, 0)$ at altitude $z_d = 10$ m. The GLASS shaping gain is set to $k_G = 0.08$ (CCW), with a non-ideal course channel $(k_\chi, \omega_{\max}) = (3.0, 0.8)$ and filtering $\tau_{\chi_d} = 0.6$ s. The translational outer loop uses $(V_{\text{ref}}, k_v, a_{\max}, k_{\text{rad}}) = (2, 2.5, 8, 0.8)$ and tilt limits $|\phi_d|, |\theta_d| \leq 40^\circ$.

Figures 7 and 8 illustrate the 6DOF simulation results. As shown in Fig. 7, the UAV captures the desired circular standoff orbit from outside initial position. In Fig. 8, the position tracking plots confirm that $(x(t), y(t))$ converge to the sinusoidal reference generated by the phase-locked orbit, while the altitude $z(t)$ stabilizes rapidly at $z_d = 10$ m. The attitude responses (ϕ, θ, ψ) exhibit damped oscillatory transients during the capture phase and then settle smoothly, indicating adequate inner-loop bandwidth relative to the outer-loop guidance dynamics. The control input profiles shows an initial thrust increase to achieve climb and orbit capture, followed by convergence of U_1 toward the steady hover-plus-centripetal requirement, while the moment (U_2, U_3, U_4) decay to near-zero mean values once steady orbiting is achieved, confirming that sustained turning is realized primarily through constant tilt rather than persistent torque excitation.

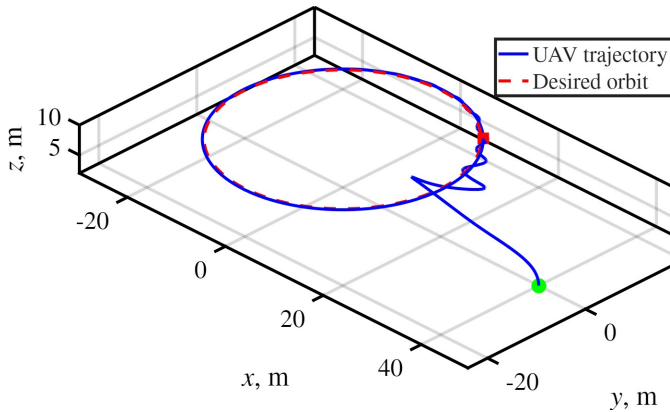


Fig. 7: UAV trajectory

V. CONCLUSIONS

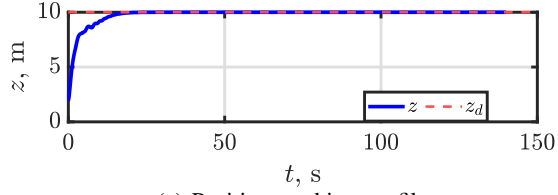
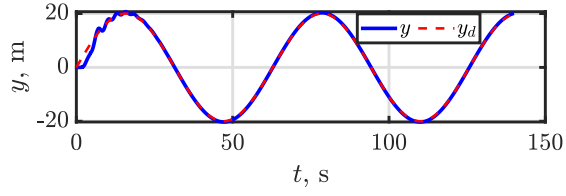
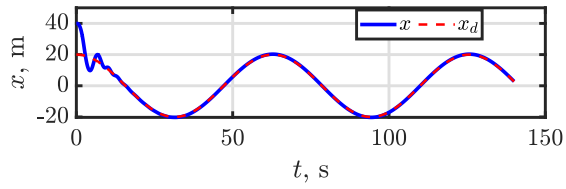
This paper introduced a GLASS guidance framework for standoff target tracking under turn-rate constraints. In contrast to gain-amplified shaping strategies, where feasible convergence speed is limited by initial-condition-dependent gain bounds, the proposed approach regulates radial contraction through a nonlinear look-angle shaping function.

The engagement dynamics were formulated in polar coordinates, enabling analytical settling-time characterization under radial-dominant conditions. Lyapunov analysis established global asymptotic convergence to the prescribed standoff circle while embedding curvature feasibility directly within the guidance design. Simulation results demonstrated faster far-field capture without excessive turn-rate demand, thereby expanding the operational envelope for practical surveillance and inspection missions.

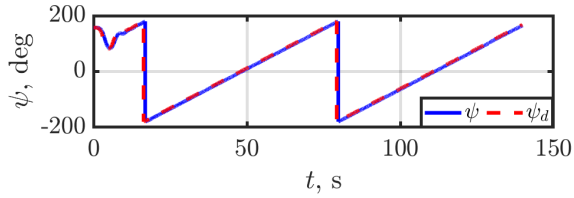
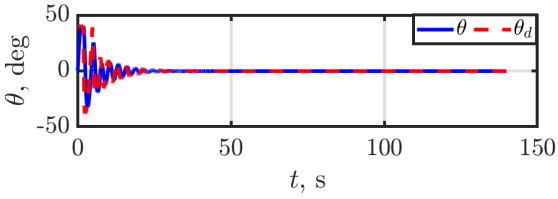
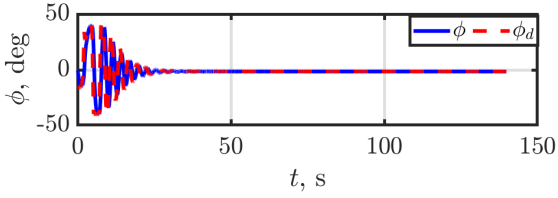
Future work will address moving targets and three-dimensional extensions, with outdoor experimental tests.

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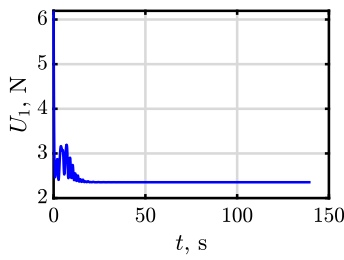
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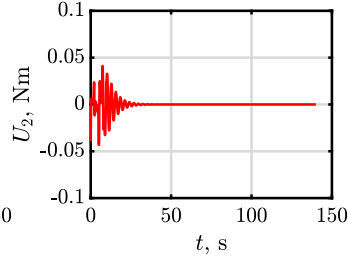
(a) Position tracking profiles



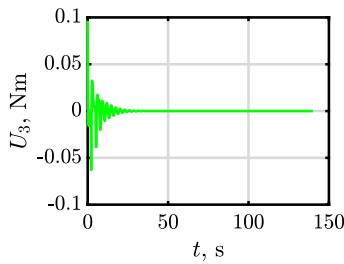
(b) Attitude tracking profiles



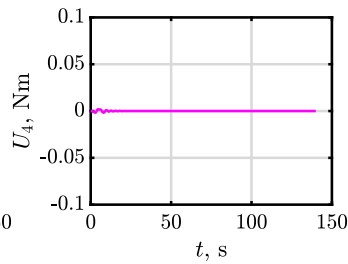
(c) Thrust profile



(d) Roll profile



(e) Pitch profile



(f) Yaw profile

Fig. 8: Results for 6dof dynamic model